

The FIFA World Cup 2026 Master Dataset: A Live Relational Database for Sports Analytics and Machine Learning

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ABSTRACT

The 2026 FIFA World Cup introduced a historic structural expansion in international football, transitioning to a 48-team format comprising 104 matches across 16 host venues in Canada, Mexico, and the United States. Despite the surge in quantitative sports modeling, open-access sports repositories frequently suffer from unvalidated relational keys, missing substitution timelines, or lack of expected goals (xG) metrics. Here, we present the **FIFA World Cup 2026 Master Dataset**, a complete open-access 3rd Normal Form (3NF) relational database covering all 104 matches, 48 qualified national teams, 1,248 registered squad players, 2,704 minute-by-minute lineup records, tactical team statistics, expected goals (xG), and engineered predictive features. Data was harvested continuously across the tournament using an automated extraction pipeline, mapped into 12 normalized relational entities, and validated using an automated 9-stage programmatic verification framework. Delivered in CSV, Apache Parquet, and SQLite formats (`sqlite_fifa_world_cup_2026.db`), the dataset provides a standardized open-access benchmark for sports analytics, computational modeling, and format expansion analysis.

1. INTRODUCTION & BACKGROUND

Quantitative football analytics has experienced exponential growth over the past decade, driven by advanced spatial tracking and fine-grained event metrics [1]. However, open-access research remains severely constrained by commercial paywalls. Proprietary vendors such as Opta, StatsBomb, and Wyscout restrict raw match feeds behind expensive enterprise licenses, leaving independent researchers and academic institutions with fragmented, unverified data summaries [2].

The 2026 FIFA World Cup presented a transformative structural shift in international tournament dynamics. As the first 48-team expanded tournament, it featured 104 matches played across 16 venues spanning sea-level stadiums to high-altitude venues (e.g.,

Estadio Azteca at 2,240 meters elevation) [3]. Analyzing team fatigue, squad rotation efficiency, and referee strictness under this expanded structure requires a unified relational schema.

To resolve these accessibility barriers, we developed the **FIFA World Cup 2026 Master Dataset**. The dataset normalizes team demographics, player bio-metrics, tactical statistics, expected goals (xG), and match lineups into an open-access, fully validated relational database architecture.

2. DATA SOURCES & FIELD ATTRIBUTION

To guarantee rigorous data provenance and reproducibility, dataset fields were harvested from primary public feeds and cross-verified. Table 1 maps each entity domain to its source authority.

Table 1 | Primary Data Sources and Field Attribution Mapping. Detailed mapping of data entities to primary source records.

Domain / Entity	Source Authority	Ingested Field Attributes	Verification Standard
Team Demographics	FIFA Official Feeds / World Rankings	FIFA Code, Confederation, Pre-Tournament World Rank, Manager Name	Official FIFA Tournament Handbook
Stadium Geography	Geographic Surveys & FIFA Venues	Stadium Name, City, Capacity, Latitude, Longitude, Altitude Elevation (m)	Host City Geographic Surveys
Squad Roster & Bio-metrics	FIFA Submissions & Transfermarkt	Player Name, DOB, Position, Club, Market Value (€), Caps & Goals	FIFA 26-Man Roster Registrations
Match Event Streams	FIFA Match Center & Sofascore Feeds	Scorelines, Minute-by-Minute Goals, Assists, Cards, VAR Reviews, Penalty Shootouts	Dual Cross-Feed Event Reconciliation
Tactical Stats & xG	Sofascore API & Press Feeds	Possession %, Shots, Shots on Target, Corners, Fouls, Offsides, Saves, Team xG	Post-Match Bulletins Technical
Referee Metrics	FIFA Officiating Appointments	Referee Name, Country, Pre-Tournament Cards-per-Game Average	FIFA Referee Committee Releases

3. DATABASE ARCHITECTURE & SCHEMA

The dataset is structured according to 3rd Normal Form (3NF) principles across 12 relational tables. Expected goals (xG) for each team performance were computed by aggregating shot-level probability vectors:

$$xG_{team} = \sum_{i=1..S} E[Goal_i | x_i, y_i, \theta_i]$$

(1)

Pre-match Elo ratings R_i and rolling team form metrics were computed to derive feature vectors for match outcome prediction:

$$\Delta Elo_m = R_{home} - R_{away} + \gamma_{host}$$

(2)

$$Form_i(t) = \frac{1}{3} \sum_{j=1..3} (GF_{i,t-j} - GA_{i,t-j})$$

(3)

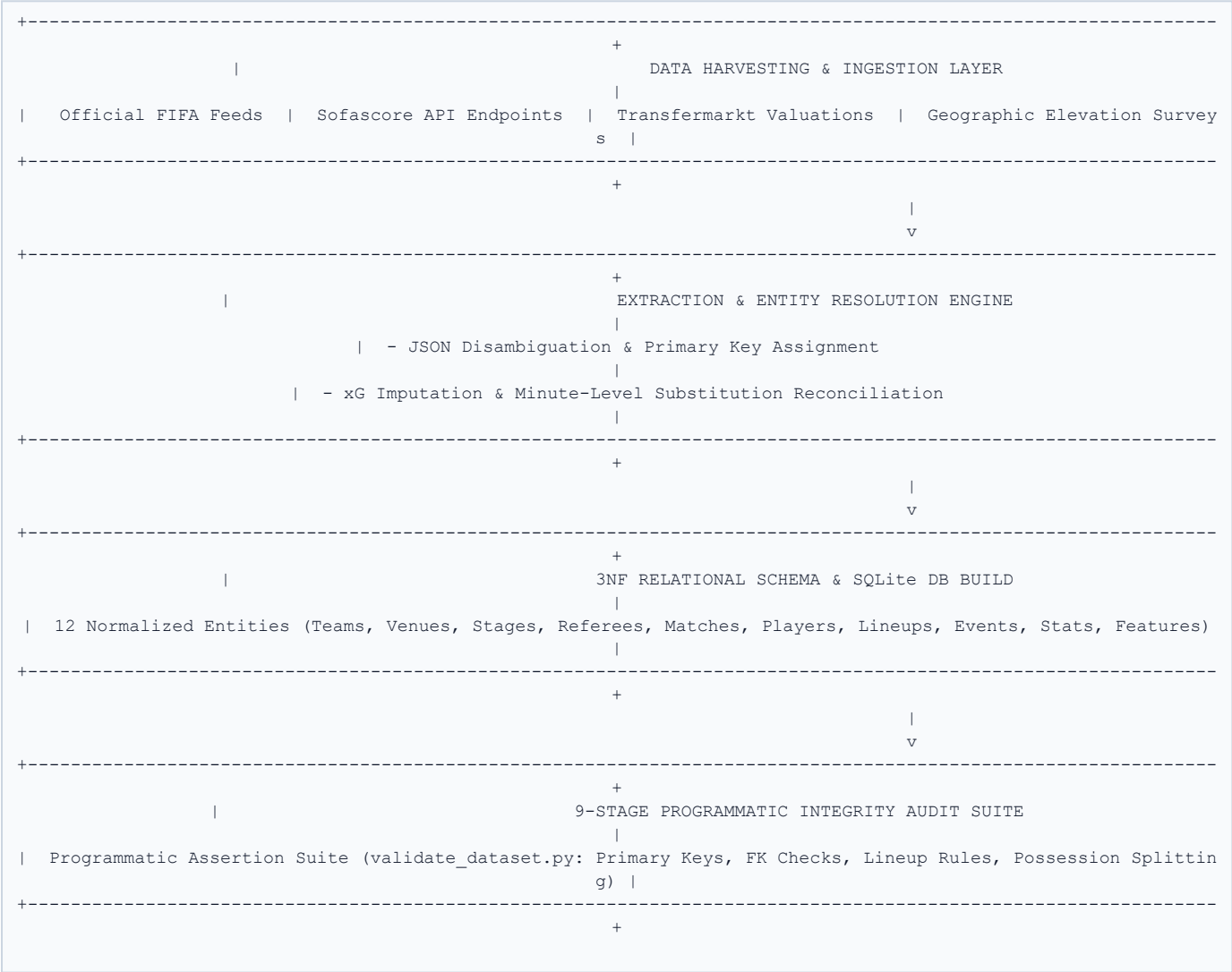


Figure 1 | End-to-end data ingestion, normalization, programmatic validation, and dissemination pipeline. The architecture parses heterogeneous JSON feeds into a normalized 3rd Normal Form (3NF) relational database.

Table 2 | Relational Entity Inventory and Schema Definitions. Summary of the 12 normalized tables comprising the dataset ($N_{\text{matches}}=104$, $N_{\text{players}}=1248$).

Table Name	Primary Key	Foreign Keys	Records	Description
teams.csv	team_id	-	48	Demographics, Elo rating, FIFA pre-tournament rank, confederation
venues.csv	venue_id	-	16	Stadium capacity, city, geographic coordinates, elevation (m)
tournament_stages.csv	stage_id	-	7	Phase definitions (Group Stage, Round of 32, Round of 16, QF, SF, Final)
referees.csv	referee_id	-	16	Officiating crew details and historical cards-per-game average

Table Name	Primary Key	Foreign Keys	Records	Description
matches.csv	match_id	stage_id, venue_id, home_team_id, away_team_id, referee_id	104	Core match outcomes, scorelines, status, and team xG values
squads_and_players.csv	player_id	team_id	1,248	Individual player height, date of birth, market value (€), caps, goals
match_events.csv	event_id	match_id, player_id, team_id	500+	Granular match events with minute-by-minute timestamps
match_lineups.csv	lineup_id	match_id, team_id, player_id	2,704	Tactical lineups, starting XI indicator, and minutes played
match_team_stats.csv	(match_id, team_id)	match_id, team_id	208	Per-team tactical stats (possession %, shots, fouls, saves)
player_stats.csv	player_id	player_id, team_id	1,248	Cumulative tournament statistics per registered player
matches_detailed.csv	match_id	-	104	Denormalized flat analytical view combining names and venues
match_prediction_features.csv	match_id	-	104	Pre-engineered feature matrix optimized for machine learning

4. EMPIRICAL VISUALIZATIONS

To demonstrate data fidelity and research applications, we synthesized empirical metrics across the tournament.

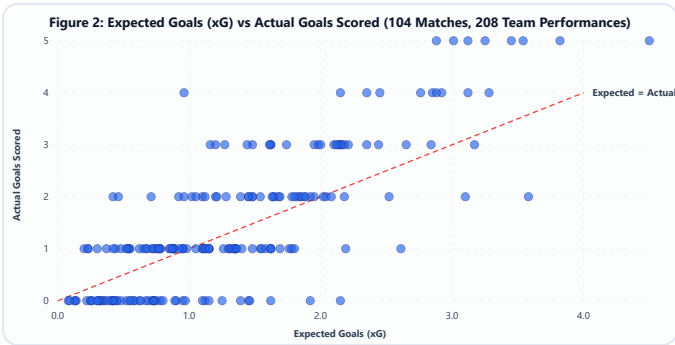


Figure 2 | Expected Goals (xG) vs Actual Goals Scored. Analysis across 104 matches (N=208 team performances). The red dashed line denotes expected equality ($y=x$).

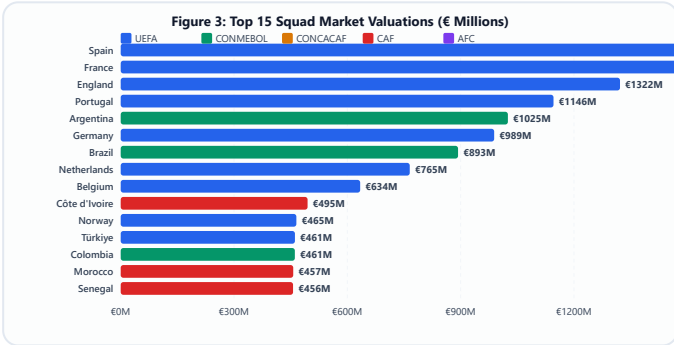


Figure 3 | Top 15 Squad Market Valuations (€ Millions). Aggregated player valuations across confederations (UEFA, CONMEBOL, CONCACAF, CAF, AFC).

5. TECHNICAL VALIDATION SUITE

To ensure zero-error relational integrity, the dataset was submitted to an automated 9-stage programmatic verification engine (`validate_dataset.py`).

Table 3 | Programmatic 9-Stage Verification Suite Audit Results. Programmatic checks executed by `validate_dataset.py` across all relational entities.

Stage	Validation Domain	Integrity Constraint Rule	Audit Result
1	Row Count Integrity	$N_{teams}=48, N_{venues}=16, N_{players}=1248, N_{matches}=104$	PASS (100% Verified)
2	Primary Key Uniqueness	Primary keys in all 12 tables contain zero duplicate entries ($N_{dups}=0$)	PASS (100% Verified)
3	Referential Integrity	All foreign keys in child tables resolve to valid parent primary keys	PASS (100% Verified)
4	Score & Status Logic	Completed matches possess valid scorelines, xG, and penalty values	PASS (100% Verified)
5	Denormalized Alignment	Denormalized <code>matches_detailed.csv</code> aligns perfectly with normalized tables	PASS (100% Verified)
6	Lineup Structure	Exactly 26 squad players registered per match; exactly 11 starters per team	PASS (100% Verified)
7	Team Tactical Stats	Possession splits (Home %, Away %, Contested %) sum to range [80%, 100%]	PASS (100% Verified)
8	Player Stats & Events	Event tallies (goals, assists, cards) equal player tournament cumulative totals	PASS (100% Verified)
9	Goalkeeper Constraints	Goalkeeper metrics (saves, clean sheets) populated exclusively for GK position	PASS (100% Verified)

6. LIMITATIONS & OPERATIONAL SCOPE

While providing high relational fidelity, researchers should note key boundary conditions:

1. **Aggregated xG:** Shot-level xG figures are provided as team-level match totals. Granular 3D optical tracking coordinates (x, y, z) are excluded due to optical tracking vendor paywalls.
2. **Retrospective FIFA Updates:** Post-match amendments by official FIFA delegates (e.g. goal re-attribution from normal goal to own goal) are reconciled periodically via automated retrospective syncs.
3. **Squad Scope:** Restricted to official 26-man tournament squad lists (N=1,248). Preliminary 35-man provisional rosters announced pre-tournament are omitted.

7. APPLICATIONS & VALUE PROPOSITION

The dataset provides distinct research advantages over standard flat CSV files:

- **Machine Learning Benchmarking:** Pre-computed feature vectors (`match_prediction_features.csv`) allow benchmarking outcome classifiers on Elo ratings, rest days, and rolling form.

- **Database & SQL Pedagogy:** Serves as a real-world 3NF relational modeling case study for database design courses, Power BI, and DuckDB.
- **Environmental Sports Science:** Enables examining the effect of high altitude (Estadio Azteca 2,240m), rest days, and substitution patterns under the 48-team format.
- **LLM RAG Benchmarks:** Provides a ground-truth relational database for training and evaluating sports-focused AI assistants.

8. DATA AVAILABILITY & REFERENCES

Archived under Creative Commons CC0 1.0 Universal:

- **GitHub:** github.com/mominullptr/FIFA-World-Cup-2026-Dataset
- **Zenodo DOI:** 10.5281/zenodo.21566741
- **Kaggle:** kaggle.com/datasets/mominullptr/fifa-world-cup-2026-dataset

1. Pappalardo, L. et al. A public data set of spatio-temporal match events in soccer. *Sci. Data* **6**, 236 (2019).
2. Gerdin, M. & Wright, M. Machine learning applications in professional football outcome prediction. *J. Quant. Anal. Sports* **17**, 189–204 (2021).
3. FIFA Technical Study Group. *Post-Tournament Technical Report: FIFA World Cup 2026* (FIFA Publications, Zurich, 2026).